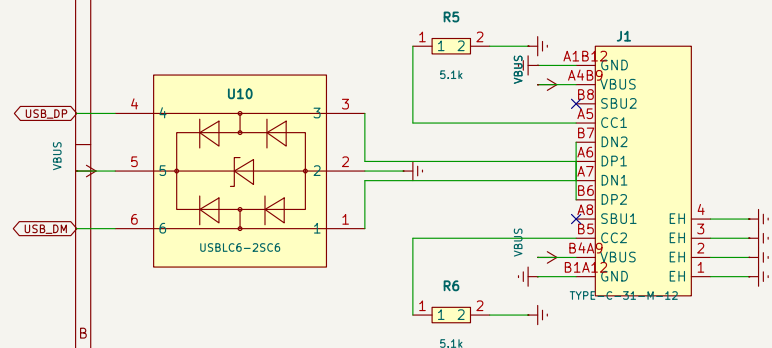
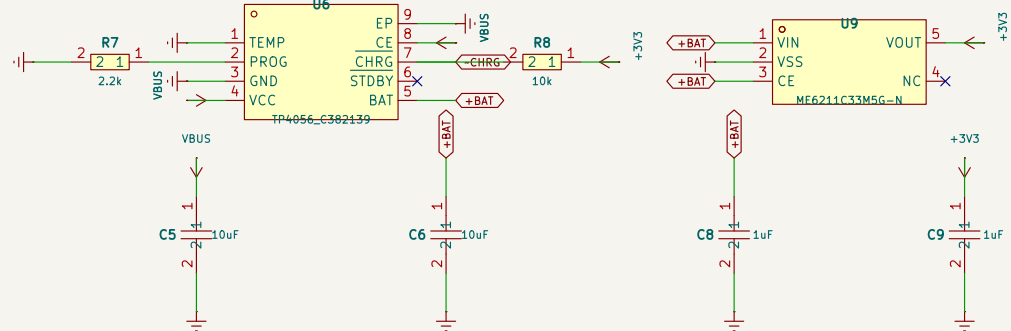


USB-C input, ESD, and charger



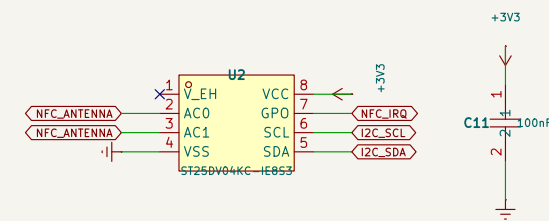
J1/U10 support: R5/R6 CC pull-downs; C5 USB input bulk

3.3 V regulation and fuel gauge

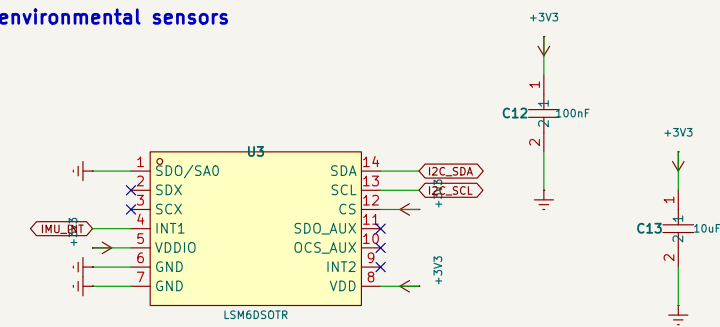


U6 support: R7 charge-current set; R8 CHRG pull-up; C5/C6 bulk

NFC dynamic tag

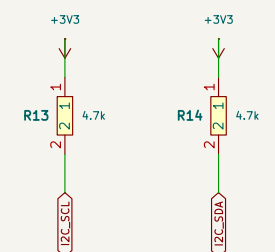


Motion and environmental sensors

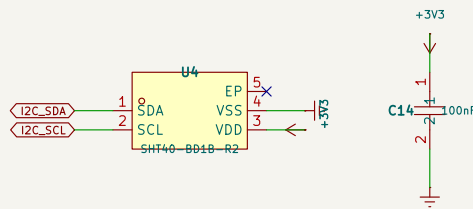


C12/C13 -> U3: IMU bypass + bulk

I2C pull-ups

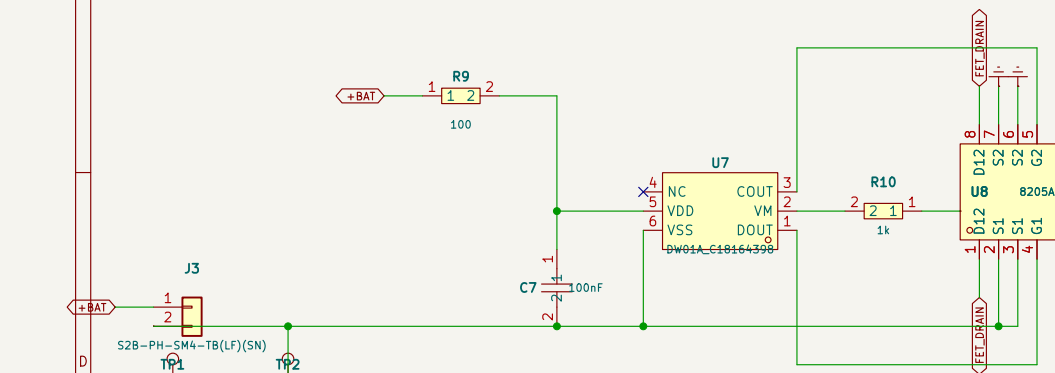


R13/R14 -> I2C bus U1-U5: SCL/SDA pull-ups



C14 -> U4: environmental-sensor bypass

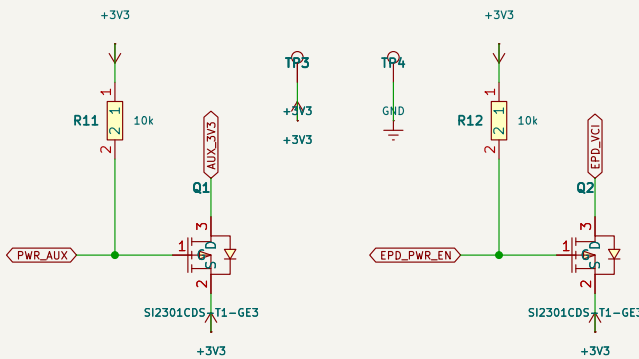
Cell protection



U7/U8 support; R9 supply feed; R10 pack sense; C7 bypass

Switched peripheral rails

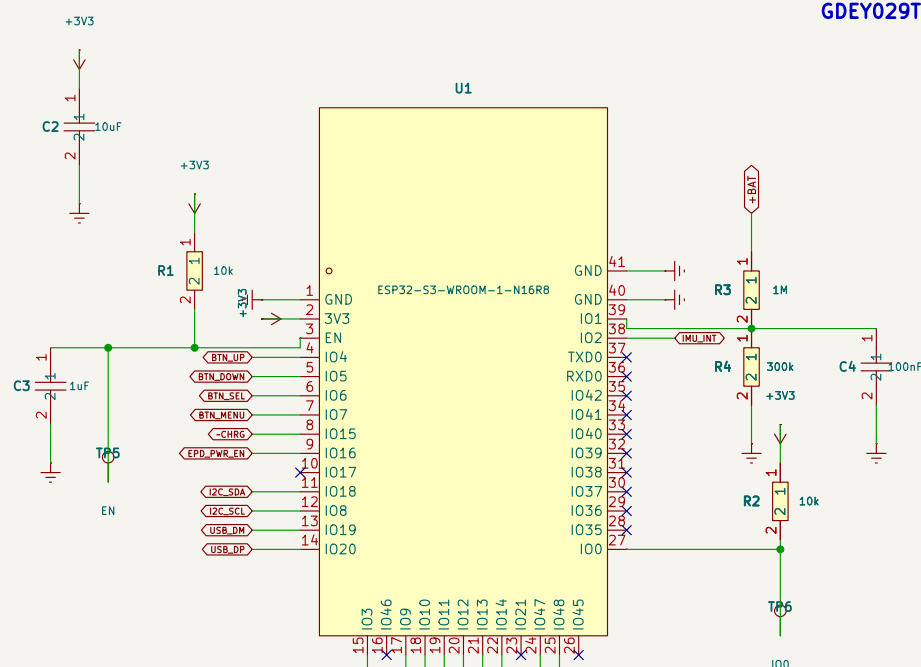
Q1/Q2 support: R11/R12 gate pull-ups (rails default off)



Controller, boot straps, decoupling, and battery ADC

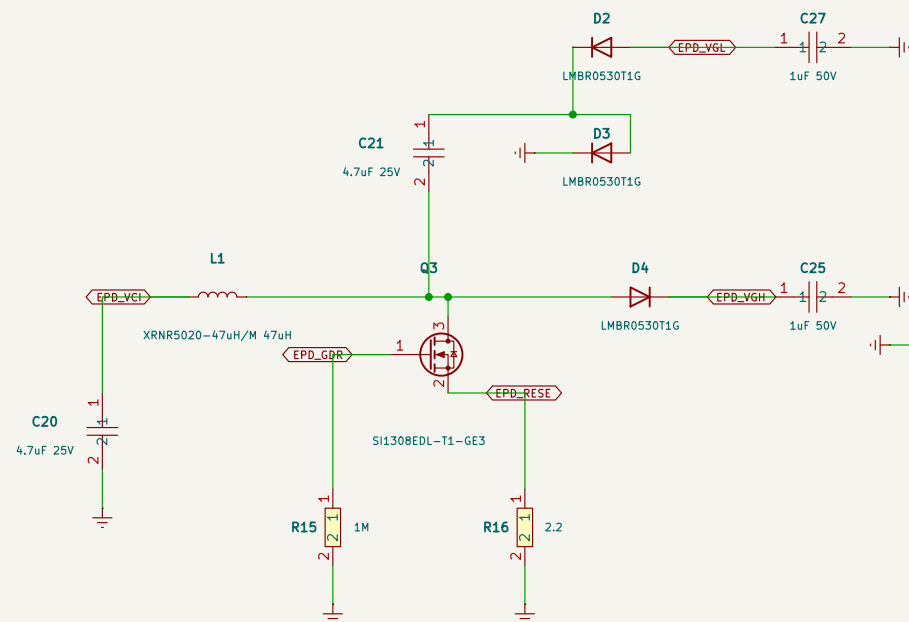
Named labels keep long inter-block connections readable.

C1/C2 -> U1: 3V3 bypass + bulk; R1/C3 -> U1 EN: pull-up + startup timing



GDEY029T94 24-pin FPC interface

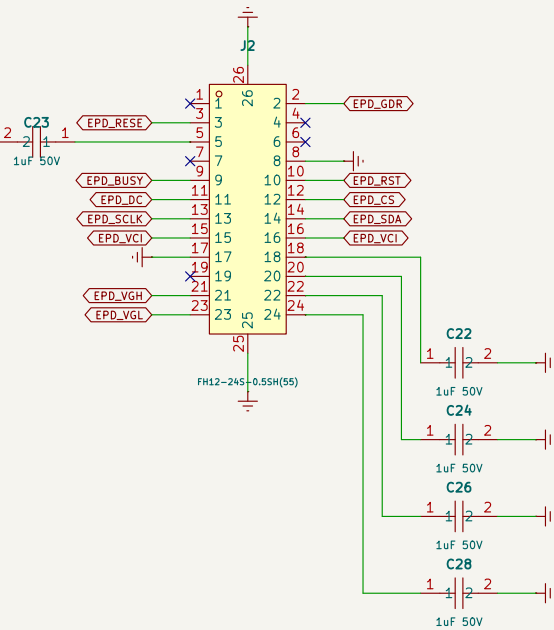
Booster and boosted gate rails



C20/L1: VCI input; C21/D2–D4/Q3: boost conversion

R15/R16: Q3 gate pull-down and current sense

J2 panel connector and direct rail reservoirs



C22-C24/C26/C28: direct J2 rail reservoirs

C25/C27: boosted VGH/VGL reservoirs

Connectivity generated from circuit.py

Single-page functional schematic

the-card

Sheet: /

File: the-card.kicad_sch
Title: the-card - ESP32-S3 e-paper smart badge

Size: A2

KiCad E.D.A. 10.0.5

10

Rev: A

Id: 1/1